BMC Health Services Research

RESEARCH Open Access

Infectious disease surveillance in Dutch hospitals: What's in it for us?

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Abstract

Background The COVID-19 pandemic has shown the need for multipurpose and sustainable hospital surveillance data. We studied the barriers, facilitators, and preferred requirements for an infectious disease surveillance system in Dutch hospitals to be valuable and feasible from a hospital perspective.

Methods This study is part of a larger EU4Health Project, UNITED4Surveillance. Semi-structured interviews with key stakeholders were conducted, and were subsequently analyzed using a thematic analytic approach.

Results Participants discussed surveillance systems for antimicrobial resistance, healthcare-associated infections, and severe acute respiratory infections. Perceived barriers to establishing or maintaining such systems included unclear and/or passive hospital roles, legal challenges, and insufficient resources and capacity. Perceived facilitators included clinically relevant and actionable data, mutually beneficial collaborations between stakeholders, and sufficient support at multiple decision-making levels and across legal, technical, capacity, and financial domains.

Conclusions This qualitative research provides new insights into the barriers and facilitators in infectious disease surveillance in Dutch hospitals from the hospital perspective since the end of the COVID-19 pandemic. Key stakeholders outlined practical recommendations to improve robust and sustainable infectious disease surveillance in Dutch hospitals: (i) ensure clinically relevant actionable data; (ii) clarify roles and responsibilities; and (iii) collaborate on innovative initiatives.

Keywords Hospital, Surveillance, Infectious disease, Semi-structured interviews

Background

The COVID-19 pandemic has shown the urgent need for robust and sustainable hospital data that can serve multiple purposes, such as providing input for public health surveillance, individual patient care, hospital bed capacity planning, and healthcare policy. Antimicrobial resistance (AMR) surveillance (ISIS-AR) [1] and healthcare-associated surveillance (PREZIES) [2] in hospitals have proven to be robust and sustainable in the Netherlands. However, over the last decade, the National Institute for Public Health and the Environment (RIVM) has initiated several attempts to set up a severe acute respiratory infections (SARI) surveillance system in the Netherlands [3], including SARI surveillance based on the data from a quality-of-care registry in intensive care units (ICUs) [4] and SARI surveillance based on financial claim codes [5]. The reasons for not achieving a sustainable SARI surveillance system yet are multifactorial and include a high registration burden, lack of fully automated and timely data transfer, a large diverse group of stakeholders, and varying appreciation of the value of epidemiological surveillance data [6]. During the COVID-19 pandemic, ad-hoc

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crisis legislation came into force, classifying COVID-19 as a notifiable disease and enabling national SARS-CoV-2 surveillance in patients admitted to ICUs and hospital wards. After the crisis legislation expired, SARS-CoV-2 surveillance was no longer sustainable in its existing form due to, among others, loss of its notifiable disease status, insufficient automatization, and the high burden of manual reporting on medical personnel. Nonetheless, establishing robust SARI surveillance in the Netherlands remains important. Since the impact of the COVID-19 pandemic might have shifted the opinion of stakeholders regarding the value of infectious disease surveillance based on hospital data and their level of engagement in pandemic preparedness, evaluating the gaps and needs with regard to hospital surveillance from the perspective of the most relevant stakeholders is needed. Therefore, we performed qualitative research using semi-structured interviews to provide an overview of the barriers, facilitators, and preferred requirements for an infectious disease surveillance system in hospitals to be valuable and feasible from a hospital perspective.

Methods

Design

We conducted semi-structured interviews with key stakeholders regarding infectious disease surveillance in hospitals in the Netherlands. This qualitative research is part of a Dutch pilot study within a larger EU4Health Project, UNITED4Surveillance (Work package 3: hospital surveillance) [7]. The Joint Action UNITED4Surveillance (project period 2023–2026), consisting of 24 Member States, aims to strengthen infectious disease surveillance for improved preparedness against cross-border health threats in Europe [8]. Since sharing study designs and best practices among Member States is an essential part of the Joint Action UNITED4Surveillance, other countries can replicate our pilot study in the future.

Participants

The most relevant stakeholder groups (i.e., organizations and their representatives) with moderate to high influence and interest in hospital surveillance were identified in a stakeholder analysis that was systematically performed *prior* to the current study. This stakeholder analysis was conducted during the period 2023-24 by RIVM colleagues and consisted of defining the role (positioned in a role-matrix), as well as the power and interest of each stakeholder (positioned in a Mendelow's matrix), and a force-field analysis on individual level [9]. Within these stakeholder groups we used a purposive sampling approach to select key stakeholders based on variation in job position, type of hospital (academic/general), and/or medical specialty focus. These stakeholders included clinical staff from different departments in hospitals; clinical staff from microbiological laboratories; members of hospital boards; policy officers from different departments within the Ministry of Health, Welfare and Sport; and representatives from medical specialist associations in the Netherlands. The selected key stakeholders were invited by email to participate in a semi-structured interview on infectious disease surveillance in hospitals. Prior to the interview, all participants received an information letter about the interview study and an informed consent form. Written informed consent was obtained from all participants and were securely stored at the RIVM for a period of 10 years (for more details see: Declarations - Ethics approval and consent to participate).

Interviews

The interviews took place using videoconferencing from October 2023 through January 2024. A semi-structured interview guide was established with a predefined topic list, which was developed by the authors and two additional researchers from the Centre for Infectious Disease Epidemiology and Surveillance of the RIVM (for details see Supplementary material 1). The guide included open-ended questions on functional infectious disease surveillance systems in hospitals, existing barriers and facilitating factors, available databases/sets/standards for potential use, and the preferred requirements for hospital surveillance from a hospital perspective. The interviews were conducted in Dutch by two RIVM researchers: a senior epidemiologist (TB) and an infectious disease specialist (SDM). The role of interviewer or note taker was divided prior to each interview between these two researchers. The interview duration was approximately 45 min, ranging from a half hour to one hour. On two occasions out of fourteen interviews, group interviews took place on specific request by the invited stakeholder group. All the other interviews were held with a single participant. All interviews were video-recorded and subsequently converted into audio files to facilitate anonymous transcription by an external company (Uitgetypt.nl). The verbatim transcripts were shared with the participants for verification. The interview quotations used for this article were translated from Dutch to English by the first RIVM researcher, and the translation was checked by the second RIVM researcher.

Analysis

The two interviewers independently identified and coded segments within each transcript using the qualitative data analysis software (MAXQDA, version 24.1.0). Codes close to the topic list were used as a starting point for overarching themes, and for the underlying (sub) themes codes were used as they emerged from the data. If the coding of the researchers diverged, the assessment was re-evaluated to reach consensus. Using a thematic

analytic approach, we identified common themes and their associations. When sufficient data saturation was reached to draw conclusions from thematic analysis, i.e., no more new codes were identified, qualitative data collection was stopped.

Results

We conducted twelve one-on-one interviews and two group interviews, involving in total 17 stakeholders. The professions of the participants in interviews were medical microbiologists, infectious disease specialists, intensivist, elderly care physician, hospital board director, clinical epidemiologist, and infectious disease pediatricians. The participants of the first group interview included three policy officers: one in hospital care, one in long-term care, and one in COVID-19. A policy officer and medical microbiologist took part in the second group interview (Supplementary material 2, Table 1).

Operational infectious disease surveillance systems in hospitals

In general, the participants mostly discussed three large groups of operational infectious disease surveillance systems in hospitals: (i) antimicrobial resistance (AMR) surveillance systems, (ii) healthcare-associated infection surveillance systems, and (iii) severe acute respiratory infection (SARI) surveillance systems. Antimicrobial resistance surveillance systems were often described by participants in relation to outbreak detection. The participants described healthcare-associated infection surveillance systems, such as postoperative wound and line infections, more frequently in relation to 'infection prevention control' and 'quality-of-care'. In general, most participants perceived AMR surveillance and healthcare-associated infection surveillance (e.g., line infection and postoperative wound infection) [10] as useful and clinically relevant. The operational respiratory infection surveillance systems in hospitals discussed by the participants were characterized by their large diversity, ranging from (inter)personal observations to fully automated, (near) real-time laboratory surveillance of respiratory pathogens. The perceived value of a future SARI surveillance system in their hospital by hospital-based participants is mixed. Some participants indicated that the value of a future SARI surveillance system might be regarded as less relevant by medical specialties other than medical microbiology or infectious disease. Other participants in the hospital setting underlined the relevance of a robust SARI surveillance system, particularly during outbreaks.

Barriers

Unclear stakeholder roles and responsibilities

A common theme apparent from the interviews was the unclear or passive role of hospitals in public health surveillance (Supplementary material 3, Table 2). Several medical specialists expressed that they do not feel that hospitals have a distinct role and/or responsibility in public health surveillance. Although most participants found it important to be notified if an infectious disease outbreak with an impact on hospital care is detected:

> *If there is something unusual we need to know, I assume we will be told. There are plenty of channels where you can find that information. So if it is necessary, that it is there. Of course, it is also a matter of trust. I assume - I think most doctors have this trust - that the RIVM will report issues if there is something noteworthy. (participant 1)*

Insight into the current role, responsibilities, and perceived urgency of having a robust respiratory surveillance in place in hospitals, was reflected by the one participant as follows:

> *Every year we know that respiratory syncytial virus (RSV) will come again, and it is the same question every year: is it more? Are we testing more now? How bad is it now? [...] Then I will go on the news and say: we are on the edge of being overwhelmed, but we will still manage [...] How many admissions there really are and whether it is worse than last year we don't know. With the Mycoplasma outbreak, the mega Mycoplasma outbreak that we had, we only found out about it, when [hospital capacity] was exceeded. (participant 13)*

Other participants expressed their willingness to actively participate in infectious disease surveillance with a broader public health scope in hospitals, but encountered problems to effectuate this. This was due to a lack of clear role descriptions and a supportive legal framework.

> *But it's things like that, and rules, that you run into. That there should be a vision about this: what is public health? And what is the role of hospitals in public health? And define that we in fact do have a role. Now, it is so strictly separated that we have no role. And that means you can't do anything. (participant 14b)*

Different interpretation of general data protection regulation

Dealing with different interpretations of the General Data Protection Regulation (GDPR) in relation to infectious disease surveillance in hospitals was regarded by many

participants as a major challenge. Several participants mentioned the tendency of local privacy officers in hospitals to be conservative and nonuniform in their advice to medical specialists about sharing data for public health reasons. According to these participants this involves both sharing and linking patient and/or laboratory data (e.g., sequence data) for public health surveillance. A participant described the legal challenges experienced by hospitals in the following example:

> *GDPR, yes. The interpretation of that law is also very dependent on the lawyer who looks at it or the institute where you work. Of course, that's all complicated. It [privacy issues] is maybe an even bigger threat, soon with AI, than the lack of support. We do monitor and share data on infectious diseases, but it has now gone so far that, for example, for healthcare evaluation, where the law simply states that this is allowed, that some lawyers still say: "don't do it." That's an issue. (participant 13)*

In addition, there also seems to be a lack of knowledge or expertise in hospitals on what is allowed under the GDPR regarding data sharing for public health in case of an infectious disease outbreak, as one participant explained:

> *[…] To what extent are you allowed to exchange data according to the GDPR? So far no one can tell me that, except, "[phew], very difficult" and "I don't know if it's allowed. (participant 4)*

Insufficient resources and capacity

An important barrier in infectious disease surveillance in hospitals is related to insufficient information and communications technology (ICT) capacity:

> *Therefore, more people in the team, or more ICT people in such a team are needed, […] That is really the bottleneck often when things don't get off the ground. And that applies to me with my team, but I also think that all other parties that work with surveillance data, and automate it or generate overviews, simply [rely] on ICT capacity. (participant 2).*

According to several participants, ICT personnel who also have a good understanding of the medical issues at hand are needed. Ideally, they could assist medical staff in translating problems encountered in daily clinical care into ICT solutions. In the setting of limited ICT capacity and funding in many hospitals, prioritization of projects by decision-makers in hospitals becomes inevitable. Top-down prioritization of infectious disease surveillance within a hospital would then be beneficial according to some participants. One participant explained why an operational SARI surveillance system has not yet been established in their hospital:

> *But also the priorities and workload in the ICT department mean that, I think, it has not yet been successful. So that's workload but also prioritization. If it was considered important enough, I think it might have been possible sooner. (participant 6)*

Facilitators

Clinically relevant and actionable data

A prominent theme regarding challenges in infectious disease surveillance in hospitals was related to the clinical relevance of these hospital surveillance activities (Supplementary material 3, Table 2). Willingness to participate in infectious disease surveillance seemed related to whether the collected data were clinically relevant and actionable enough for patient care to have added value for hospitals, which was considered to have direct consequences for medical treatment and/or medical policy in hospitalized patients. In this context, many participants expressed that infectious disease surveillance in hospitals becomes worthwhile if it improves patient quality of care and/or facilitates healthcare innovation. The positive returns of establishing infectious disease surveillance in hospitals have become increasingly clear to participants in recent years, such as regional AMR surveillance in the following example:

> *Because you continuously perform surveillance, we can act very quickly when it comes to infection prevention, and the lines of communication are also extremely short. I do that for this hospital, but my colleagues do it for the other hospital and we can all view those results. As a result, there are few intermediate links. (participant 4)*

In the context of a demanding hospital setting with a high administrative burden and staff shortages, it was also regarded as essential to have a critical attitude about data requests by external stakeholders for infectious disease surveillance in hospitals in general, as was described by a participant:

> *Therefore, with every data source you ask for, I think you have to ask yourself: do I really need it? But also: where can you find these [data]? Is there already a standardization for this? And if not, then you really have to ask yourself whether that is something you should want to collect on a large scale in this phase, […] (participant 12).*

Mutually beneficial collaboration

One of the most important facilitators for establishing robust infectious disease surveillance in hospitals appeared to be mutually beneficial collaboration with relevant stakeholders. For instance, a participant described the close collaboration between a large academic hospital and the regional municipal health service within the Amsterdam Regional Genomic epidemiology and Outbreak Surveillance (ARGOS) consortium as illustrative of the advantages of sharing surveillance data and expertise, which could lead to direct action at the regional or local level. Other long-term collaborations that have intensified between regional microbiology laboratories and hospitals since the COVID-19 pandemic are illustrated by one of the participants as follows, regarding respiratory pathogens, such as influenza and SARS-CoV-2:

> *[…] And from this a regional assembly originated in which all medical microbiologists and infection prevention experts are represented, and more and more regional policy decisions are being made. So, the great thing about that is: you share information, you share policy, and you also share in decisions, and I think that is very powerful and very nice that it has come about that way. That is also getting better and better. (participant 9)*

Sufficient support

A sufficient degree of support was described as valuable for facilitating robust and sustainable infectious disease surveillance in hospitals. This support was required on multiple levels, ranging from supportive legislation and expertise, a blueprint for implementing hospital surveillance, sufficient funding for diagnostics with a public health aim, to prioritization by both the medical staff and the medical board. With regard to the preferred support, several participants noted that mandatory infectious disease surveillance in hospitals, excluding the already notifiable diseases, is not regarded as useful, but a (legal) framework defining roles and responsibilities is.

Recommendations

The participants described strategies on how the most important challenges could be faced in establishing or maintaining robust infectious disease surveillance in hospitals, and on how to create added value from the hospital perspective.

Establish the roles and responsibilities of stakeholders

A common notion was that the objectives, roles, and responsibilities for infectious disease surveillance in hospitals that have a broader public health scope must be clarified to create a shared vision. A first step that a participant suggested was to have open discussions about it with the most relevant stakeholders in hospital surveillance with a public health scope from multiple levels:

> *[…] So perhaps, the solution is to look at this in a more nuanced way. What are the objectives of those types of surveillance, and can we simply explain it more clearly to each other and have a dialogue about it? (participant 11)*

In addition, several participants recommend that guidance from the Dutch Ministry of Health is also required to establish the (legal) framework for achieving sustainable infectious disease surveillance in hospitals.

> *Frameworks. Look, you just have to indicate that there are also a number of stakeholders who have to find solutions together within certain frameworks. And those frameworks must be set. All from the same money, I always think. Not an easy process, but I think [it is] one that needs to be managed if you want to get out of this. Certainly, also for the future. (participant 11)*

Ensure actionable data with clinical relevance for hospital patient care

In general, it was emphasized to focus on obtaining *actionable* surveillance data from the hospital perspective, i.e., with impact on patient clinical care to ensure the sustainability of infectious disease surveillance in hospitals. The most often cited key attributes of these hospital surveillance systems were 'timeliness' and 'flexibility.' According to participants, the required timeliness differs for specific infectious disease syndromes or pathogens under surveillance, ranging from (near) real-time to monthly to quarterly to half-yearly. Additionally, it is dependent on the surveillance objective set by the hospital and/or the current epidemiological situation. In general, the value of sufficient timely hospital surveillance data was expressed as follows:

> *It is of course information for action. So, you don't want to find out only a year later that there was a problem a year ago. So, timeliness is very important. (participant 2)*

The flexibility of surveillance systems was also regarded as an important attribute as the hospitals expressed a strong need to swiftly adapt to changing information needs. Adaptively upscaling or downscaling surveillance activities depending on the changing epidemiological situation was perceived as a requirement, especially in the setting of hospital outbreaks. In this context, a clear preference was expressed for routinely and continuously

acquired regional infectious disease surveillance data in addition to national infectious disease data:

> *For the future, it is super important to know - not only in the province, but maybe even in a part of the province - where is your problem? In which village or which…? And that you can really… As long as you can share that data with each other, you can really map it out fantastically and take specific measures. (participant 9).*

From the perspective of hospitals, having insight into infectious disease surveillance outputs at the regional level is regarded as highly beneficial, because it provides actionable data for *their* own catchment population. Especially, since regional transfers of patients occur frequently between hospitals in the Netherlands, insight into transmission dynamics at the regional level is regarded of the utmost importance. Participants elaborated on numerous initiatives (e.g., ARGOS, AMR Healthcare Networks) undertaken at the regional level by collaborating hospitals, microbiological laboratories and/or municipal health services to facilitate this need.

Benefit from the innovative qualities of hospitals

Several participants saw opportunities for establishing robust infectious disease surveillance in hospitals with a wider public health scope. One participant advised to benefit broadly from the innovative power of academic hospitals, ranging from utilizing their crosslinked networks, spin-off private companies, and technological innovations, such as artificial intelligence, to their extensive expertise with complex patient groups:

> *And I think you also should really use the innovative power of the University Medical Centers in the field of infectious diseases. That is where the University Medical Centers really have their added value. […] (participant 11).*

This innovative power of clinical microbiological laboratories was also underlined by another participant:

> *If we want to have a system in which we can respond quickly to outbreaks, then you will need a system that embraces technological innovations. And that can actually be incorporated into the whole of how techniques, lab options, those developments are used in the microbiological world, so that they are immediately passed on throughout the chain. (participant 14b).*

Discussion

To identify common ground between stakeholders, such as clinical staff from hospitals and/or laboratories, and public health institutes, it is essential that their different perspectives on the wants, needs, and requirements of hospital surveillance are recognized. To our knowledge, this qualitive research is one of the few recent Dutch studies [11, 12] since the end of the COVID-19 pandemic that provides insight into the barriers, facilitators, and preferred requirements of infectious disease surveillance from the perspective of Dutch hospitals. Within the hospital setting, this qualitative research indicated that there are nuanced differences in the perspectives on hospital surveillance depending on the aim of the syndrome/disease/pathogen under surveillance and stakeholder type. Based on the semi-structured interview results, we provide practical recommendations that might help to find common ground between relevant stakeholders to facilitate robust and sustainable hospital surveillance in the Netherlands.

The three main recommendations for maintaining or establishing robust and sustainable hospital surveillance are: (i) ensure clinically relevant actionable data, (ii) clarify roles and responsibilities, and (iii) collaborate on innovative initiatives.

First, we recommend public health institutes that collaborate with hospitals to focus on clinically relevant actionable data from the perspective of hospitals when establishing infectious disease surveillance. This approach will likely result in improved intrinsic motivation of hospitals to participate in infectious disease surveillance activities and ensure long-term sustainability of the surveillance system. Hereby, it is also essential to provide timely mirror data for feedback or benchmarking on a user-friendly, easy-to-access platform to facilitate the reporting of these surveillance data to hospital staff. In addition, the added value of these surveillance results for hospitals would increase if the geographical scale of the mirror data is adjustable according to the hospital needs (e.g., local/regional/national level). This would fulfill the need of many hospitals to adjust mirror data to the regional level, which provides them with the most clinically relevant actionable data.

Second, we recommend defining the roles and responsibilities of key stakeholders in hospital surveillance more clearly and collaboratively, especially regarding hospital surveillance with a broader public health aim. Currently, some stakeholders in hospitals seem unaware of or do not acknowledge their own distinct role in a routine and continuous infectious disease surveillance in hospitals with a broader public health scope. This is in line with a British qualitative study [13] exploring the views of healthcare professionals on the use of AMR surveillance data using semi-structured interviews. One of their findings

revealed that some medical professionals in secondary care seemed unaware of certain relevant data and surveillance systems already available. A way forward for the Netherlands could be to first set up informal meetings or round tables with key stakeholders to discuss surveillance objectives, needs and requirements of infectious disease surveillance systems in hospitals. This could lead to more insight into each other's needs, further exploration of common ground, and fine-tuning of a blueprint for sustainable SARI surveillance in the Netherlands. This process is already ongoing with a project in collaboration with Dutch Hospital Data (DHD), RIVM, the Netherlands Federation of University Medical Centers (NFU), and the Dutch Hospitals Association (NVZ) to establish SARI surveillance using financial claim codes and laboratory data. As several hospitals experienced the lack of a (legal) framework that could facilitate more clearly defined public health roles, it would be helpful to clarify the role of hospitals in the current adjustment of the Public Health Act (Wpg) in the Netherlands [14]. Providing a framework would help hospitals commit to public health tasks to a greater extent, acknowledge their responsibility, and contribute to providing baseline infectious disease data at the hospital level. Moreover, it would also raise the awareness in hospitals that routinely and continuously collected infectious disease surveillance data, are essential for adequate and timely detection of infectious disease outbreaks in hospitals.

Third, it is recommended for public health institutes to benefit from the innovative qualities of hospitals and microbiology laboratories by creating long-term, mutually beneficial collaborations with hospitals for establishing robust infectious disease surveillance. Mutually beneficial collaboration between these stakeholders could lie in technological innovations, such as the implementation of artificial intelligence technologies, which could be of added value for both infectious disease surveillance and improving the quality of patient care. A French study [15] examining infection control professionals' perception of newly implemented electronic surveillance software to support inpatient infection control, found that it enhanced their responsiveness, work efficiency and communication, ultimately improving quality of patient care. In addition, hospital surveillance data could be regarded as a valuable data source for innovative, clinically relevant research or for addressing policy questions in the field of infectious disease. The extensive innovation of hospital microbiological laboratories regarding whole-genomic sequencing was also recommended to be embraced by all stakeholders in the chain to fully utilize its potential.

This qualitative research has several strengths and limitations to consider. A general strength of the purposive sampling method is that it is an efficient way to select participants who are likely to contribute comprehensively and share diverse insights, because of their knowledge or experience with the topic [16, 17]. This can benefit the quality and accuracy of research findings [17, 18]. However, the level of knowledge and experience of participants on the topic in the current study was sometimes limited to general needs or possibilities within a medical specialty and did not always extend to specific knowledge about surveillance or potential surveillance systems. Yet, this could also be perceived as an important outcome of the current research, which informs future course of action with different stakeholders. Another potential limitation regarding purposive sampling is the potential influence of the researcher on the sampling process, leading to a biased sample [16]. However, this is likely to be limited because the sampling method was reinforced by the formal stakeholder analysis to identify as many relevant stakeholders as possible. Lastly, the group of participants did not include pulmonary medical specialists due to nonresponse. In addition, we did not include other potential stakeholders, such as nursing staff, administration officers, and/or financial managers, because they did not fall in our primary target group with high influence and high interest. However, despite these limitations, we believe that this study managed to capture the most relevant views and identify common themes among the most relevant key stakeholders with high power/interest, because sufficient data saturation occurred to allow us to draw conclusions from the thematic analysis.

Conclusions

This qualitative research provides new insights into barriers and facilitators experienced by Dutch hospitals in infectious disease surveillance from the hospital perspective since the end of the COVID-19 pandemic. Based on input from key stakeholders in the hospital setting, this study provides practical recommendations that can be of added value for improving robust and sustainable infectious disease surveillance in Dutch hospitals in the future.

Abbreviations

AMR	Antimicrobial resistance
DHD	Dutch Hospital Data
GDPR	General Data Protection Regulation
ICU	Intensive care unit
ICT	Information and communications technology
NFU	Netherlands Federation of University Medical Centers
NVZ	Dutch Hospitals Association
RIVM	National Institute for Public Health and the Environment
RSV	Respiratory syncytial virus
SARI	Severe acute respiratory infections
Wpg	Public Health Act in the Netherlands

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12913-026-14513-2.

Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

Acknowledgements
We would like to acknowledge the contributions of peer reviewers at the RIVM in drafting this paper into its final form.

Author contributions
Sierk Marbus (SDM) designed the study, analyzed and interpreted the data, and drafted the manuscript. Tjarda Boere (TB) designed the study, analyzed and interpreted the data, reviewed and edited the manuscript. Irene Veldhuijzen co-designed the semi-structured interview guide and reviewed the manuscript. Rianne van Gageldonk-Lafeber co-designed the semi-structured interview guide and reviewed the manuscript. All the authors read and approved the final manuscript.

Funding
This pilot study is part of the UNITED4Surveillance (Work Package 3 Hospital surveillance). UNITED4Surveillance is funded by the European Union under the EU4Health Programme (EU4H), project ID 101102070. The views and opinions expressed are those of the author(s) only and do not necessarily reflect those of the European Union or granting authority. Neither the European Union nor the granting authority can be held responsible for them.

Data availability
The datasets used and/or analysed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
The Dutch Medical Research Involving Human Subjects Act (Dutch acronym: WMO) did not apply to this study, because only pseudonymized data were used and there were no interventions. A full Data Protection Impact Assessment (DPIA) was not required, as a Privacy Quick Scan conducted by an RIVM privacy officer indicated a low level of risk.

Consent for publication
All participants in the semi-structured interviews gave consent for publication of the results of this qualitative research.

Competing interests
The authors declare no competing interests.

Received: 2 September 2024 / Accepted: 5 April 2026
Published online: 11 April 2026

References

1. RIVM. ISIS-AR. https://www.rivm.nl/isis-ar. Accessed 7 Jan 2026.
2. RIVM. PREZIES. https://www.rivm.nl/prezies. Accessed 7 Jan 2026.
3. Marbus SD, van der Hoek W, van Dissel JT, van Gageldonk-Lafeber AB. Experience of establishing severe acute respiratory surveillance in the Netherlands: Evaluation and challenges. Public Health Pract. 2020. https://doi.org/10.1016/j.puhip.2020.100014
4. NICE. National Intensive Care Evaluation. https://www.stichting-nice.nl/. Accessed 3 July 2024.
5. Niessen A. Developing a syndromic case definition for timely surveillance of severe acute respiratory infections based on financial claim codes in the Netherlands. In preparation.
6. Marbus SD. Severe acute respiratory infections, the missing link in the surveillance pyramid. Bilthoven: Leiden University; 2021.
7. UNITED4Surveillance. UNITED4Surveillance. https://united4surveillance.eu/. Accessed 3 July 2024.
8. European Commission. EU4Health and Horizon 2020 projects preparing the Union for future health crises. https://hadea.ec.europa.eu/news/eu4health-and-horizon-2020-projects-preparing-union-future-health-crises-2022-12-27_en. Accessed 3 July 2024.
9. Marbus SD, van Gageldonk-Lafeber AB. Infectious disease surveillance in Dutch hospitals: a UNITED4Surveillance (WP3) pilot study. 2026; Jan 7. https://zenodo.org/records/14699054.
10. RIVM. Modules PREZIES. https://www.rivm.nl/prezies/modules-PREZIES. Accessed 3 July 2024.
11. Hopmans TEM, Smid EA, van der Wille JC, Koek MBG, Vos MC, Geerlings SE, de Greeff SC. Trends in prevalence of healthcare-associated infections and antimicrobial use in hospitals in the Netherlands: 10 years of national point-prevalence surveys. J Hosp Infect. 2020. https://doi.org/10.1016/j.jhin.2019.10.005.
12. van der Verberk JDM, Kampstra NA, Reimes N, van Rooden SM, Hopmans TEM, Geerlings SE, de Greeff SC. Healthcare-associated infections in Dutch hospitals during the COVID-19 pandemic. Antimicrob Resist Infect Control. 2023. https://doi.org/10.1186/s13756-022-01201-z.
13. Al-Haboubi M, Trathen A, Black N, Eastmure E, Mays N. Views of health care professionals and policy-makers on the use of surveillance data to combat antimicrobial resistance. BMC Public Health. 2020. https://doi.org/10.1186/s12889-020-8383-8.
14. Kuipers EJ. Planning van de derde en overige tranches van de Wet publieke gezondheid (Wpg). https://share.google/sQXVr5cZBJp5eURCM. Accessed 3 July 2024.
15. Baudet A, Brennstuhl MJ, Lizon J, Regad M, Thilly N, Demore B, Florentin A. Perceptions of infection control professionals toward electronic surveillance software supporting inpatient infections: A mixed methods study. Int J Med Inf. 2024. https://doi.org/10.1016/j.ijmedinf.2024.105419.
16. Johnson JL, Adkins D, Chauvin S. Qualitative research in pharmacy education: a review of the quality Indicators of Rigor in Qualitative Research. Am J Pharm Educ. 2020. https://doi.org/10.5688/ajpe7120.
17. Palinkas LA, Horwitz SM, Green CA, Wisdom JP, Duan N, Hoagwood K. Purposeful Sampling for Qualitative Data Collection and Analysis in Mixed Method Implementation Research. Adm Policy Ment Health. 2015. https://doi.org/10.1007/s10488-013-0528-y.
18. Campbell S, Greenwood M, Prior S, Shearer T, Walkem K, Young S, Bywaters D, Walker K. Purposive sampling: complex or simple? Research case examples. J Res Nurs. 2020. https://doi.org/10.1177/1744987120927206.

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